Malthus Meets Romer: Population Growth Under Information Frictions

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Abstract

Households disagree about whether larger populations enhance productivity (Romerians) or diminish it (Malthusians). I embed this disagreement into a Barro-Becker fertility choice model with strategic interaction. Romerians exhibit strategic complementarity in fertility while Malthusians exhibit strategic substitutability. In Romerian worlds, both demographic selection and Bayesian learning favor Romerians, ensuring belief convergence. In Malthusian worlds, these forces conflict, generating boom-bust cycles. A planner announcing an estimate of the scale-effect parameter improves welfare when the estimate is sufficiently informative relative to belief dispersion. Using data from the General Social Survey, I document that individuals holding Malthusian beliefs report significantly lower ideal fertility and have fewer children than non-Malthusians, consistent with the model's central prediction.

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1 Introduction

How many children should a family have? The answer depends critically on beliefs about the economic consequences of population size. A parent who believes that more people generate more ideas and faster growth—in the spirit of Romer (1990)—might view an additional child as contributing to future prosperity. A parent who fears that more people strain finite resources—in the spirit of Malthus (1798)—might view the same child as hastening collective impoverishment. These beliefs directly shape fertility decisions under intergenerational altruism, and the resulting population dynamics feed back into productivity outcomes that either validate or refute the beliefs. This paper studies the consequences of this feedback loop when households hold heterogeneous beliefs about the population-productivity relationship.

I develop an overlapping generations model in which households make fertility decisions under uncertainty about a key structural parameter: the scale effect of population on productivity growth. Households observe the same aggregate data but interpret it through different priors. Some households are "Romerians" who believe that larger populations accelerate innovation and growth. Others are "Malthusians" who believe that larger populations depress living standards by diluting fixed factors. These beliefs are transmitted across generations through a combination of cultural inheritance and Bayesian updating from observed outcomes.

The analysis yields four main theoretical contributions. The first contribution concerns strategic interaction in fertility choices. I analyze the fertility game using techniques from the global games literature (Carlsson and Van Damme, 1993; Morris and Shin, 1998). Each household's optimal fertility depends on expectations about aggregate fertility, which in turn depends on the distribution of beliefs and on each household's expectation of others' expectations, and so on. I show that, provided the Romerian scale-effect belief is large enough to overcome the dilution of per capita consumption, Romerians exhibit strategic complementarity: holding beliefs fixed, a Romerian increases fertility when expecting higher aggregate fertility, because higher aggregate fertility raises expected future productivity under Romerian beliefs, increasing the return to children. Malthusians always exhibit strategic substitutability: a Malthusian decreases fertility when expecting higher aggregate fertility, because under Malthusian beliefs, higher aggregate fertility reduces expected future productivity *and* dilutes per capita consumption. This asymmetry in strategic interaction has important implications for equilibrium multiplicity, stability, and dynamics.

The second contribution concerns the role of incorrect beliefs. I distinguish between two scenarios: a "Romerian world" in which the true technology features positive scale effects, and a "Malthusian world" in which the true technology features negative scale effects. The dynamics of beliefs and population differ dramatically across these scenarios. In a Romerian world, Malthusians impose a social cost by suppressing fertility below the efficient level, but this cost is transitory. Both demographic selection (Romerians have more children) and Bayesian learning (high population correlates with high productivity, confirming Romerian beliefs) push the population toward Romerian dominance. Malthusian beliefs eventually vanish, and the economy converges to the efficient steady state.

In a Malthusian world, the picture is starkly different. Demographic selection still favors Romerians—optimists always have more children, regardless of whether their optimism is warranted. But Bayesian learning now favors Malthusians, since high population correlates with low productivity, confirming Malthusian beliefs. These opposing forces generate endogenous boom-bust cycles. During periods of moderate population, both types coexist; demographic selection gradually increases the Romerian share; aggregate fertility rises; population grows. Eventually, population exceeds carrying capacity, productivity falls, and young generations update toward Malthusian beliefs. The Romerian share falls, fertility declines, population stabilizes, and the cycle begins anew. Wrong beliefs—specifically, Romerian beliefs in a Malthusian world—are the source of instability.

The third contribution is an application to demographic history. The model provides a unified framework for understanding three stylized facts: the Malthusian trap that characterized most of human history, the demographic transition that accompanied industrialization, and the modern fertility decline in developed countries. I interpret the pre-industrial era as a Malthusian world in which cycles of the type described above played out over centuries. The Industrial Revolution represented a shift from a Malthusian to a Romerian technology, initially generating a lag in belief updating and a period of rapid population growth as Romerians thrived. The modern fertility decline may reflect either a return to Malthusian concerns (via environmental consciousness) or an appropriate response to automation and declining labor shares.

The fourth contribution concerns policy. I analyze the value of public announcements about the scale-effect parameter. Even when a social planner does not know the true parameter with certainty, announcing an estimate can improve welfare by coordinating beliefs and reducing fertility dispersion—provided the planner's signal noise is small relative to the dispersion of household beliefs. The welfare gain from announcement is positive whenever this informativeness condition holds, and the gain is increasing in the precision of the planner's information. When the planner's signal is essentially uninformative, however, announcing it can harm welfare by causing households to coordinate on noise.

Beyond the theoretical contributions, I provide empirical evidence supporting the model's central prediction that Malthusian beliefs reduce fertility. Using data from the General

Social Survey (GSS), I document robust negative correlations between Malthusian beliefs—measured by agreement with the statement that Earth cannot continue to support population growth—and both ideal and actual fertility. Individuals who hold Malthusian beliefs report ideal fertility approximately 0.27 children lower than non-Malthusians and have 0.29 fewer children, differences of roughly 11 and 16 percent relative to the respective sample means. These relationships persist after controlling for age, education, sex, and survey year, and are present across demographic groups. The fact that beliefs correlate with realized fertility outcomes, not just stated preferences, provides suggestive support for the model's mechanism: beliefs about population-productivity relationships shape fertility decisions that translate into behavior.

Related Literature

This paper connects several strands of literature: endogenous fertility and population, heterogeneous beliefs in macroeconomics, global games, and cultural transmission.

The foundational work on endogenous fertility is Becker (1960), who introduced the quantity-quality tradeoff, and Barro and Becker (1989), who embedded fertility choice in a dynastic optimization framework. Subsequent contributions have enriched this framework with human capital accumulation (Galor and Weil, 2000), mortality transitions (Kalemli-Ozcan, 2003), and intergenerational transfers (Lee, 2003). The unified growth theory of Galor (2011) synthesizes these elements to explain the demographic transition from Malthusian stagnation to modern growth. My contribution differs by introducing belief heterogeneity as a driver of fertility differentials and population dynamics, rather than focusing on the evolution of technology or human capital.

The debate over population scale effects has a long history. Malthus (1798) articulated the view that population growth outstrips resource growth, leading to immiseration. Romer (1990) and Kremer (1993) formalized the opposing view that larger populations generate more ideas, accelerating technological progress. Jones (1995) and Jones (2022) provide frameworks that nest both perspectives. Empirical work by Ashraf and Galor (2013) finds evidence for positive scale effects over very long horizons, while Voigtländer and Voth (2013) document Malthusian dynamics in pre-industrial Europe. I do not take a stand on which view is correct; rather, I study the consequences of households holding different views.

The literature on heterogeneous beliefs has grown substantially. In macroeconomics, Angeletos and La'O (2013) and Angeletos et al. (2018) examine how dispersed information affects business cycles and sentiments. Bordalo et al. (2018) develop diagnostic expectations with belief overreaction. Malmendier and Nagel (2011) document that personal experiences shape macroeconomic expectations. In finance, Simsek (2013), Geanakoplos (2010), and

Scheinkman and Xiong (2003) study asset pricing under disagreement. My paper differs by modeling disagreement about structural parameters governing long-run relationships rather than beliefs about current states or short-run forecasts.

The global games literature, pioneered by Carlsson and Van Damme (1993) and developed by Morris and Shin (1998) and Morris and Shin (2003), provides tools for analyzing games with strategic complementarities under incomplete information. I adapt these techniques to a setting with both strategic complementarities (among Romerians) and strategic substitutabilities (among Malthusians), which creates novel analytical challenges but also generates richer dynamics than standard global games applications such as currency crises (Morris and Shin, 1998) or bank runs (Goldstein and Pauzner, 2005).

The literature on cultural transmission, developed by Cavalli-Sforza and Feldman (1981), Bisin and Verdier (2001), and Bisin and Verdier (2011), provides frameworks for understanding how preferences and beliefs persist across generations. Doepke and Zilibotti (2019) apply these tools to study parenting styles. I incorporate cultural transmission as one channel through which beliefs evolve, alongside Bayesian learning from observed outcomes.

The analysis of public information in coordination games relates to work by Morris and Shin (2002) on the social value of public information. They show that in games with strategic complementarities, public information can be harmful if it crowds out private information. In my setting, the welfare implications of public announcements are more nuanced because of the heterogeneous strategic interaction: Romerians and Malthusians respond differently to common signals. I characterize conditions under which public announcements are beneficial.

Finally, my paper relates to work on population cycles in economic history. Lee (1987) and Voigtländer and Voth (2013) document demographic fluctuations in pre-industrial Europe. Clark (2007) synthesizes evidence on Malthusian dynamics before the Industrial Revolution. I provide a belief-based mechanism for such cycles that complements purely technological explanations.

The paper proceeds as follows. The remainder of this section discusses related literature. Section 2 presents the model environment, including the fertility choice problem, belief formation, and intergenerational transmission. Section 3 analyzes strategic complementarity and substitutability using global games techniques. Section 4 examines long-run dynamics under learning and demographic selection. Section 5 presents the empirical evidence on beliefs and fertility. Section 6 discusses the alignment between theoretical predictions and demographic history. Section 7 analyzes the value of public information and characterizes optimal communication policy. Section 8 concludes. All proofs are collected in the Appendix.

2 Model

I develop a discrete-time overlapping generations model with dynastic altruism. The economy is populated by a continuum of households who make fertility decisions. Households differ in their beliefs about the structural relationship between population and productivity growth.

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. Each period, a generation of adults makes economic decisions; their children become adults in the following period.

2.1.1 Production

A single final good is produced using labor according to the technology

$$Y_t = A_t N_t^{\alpha}, \tag{1}$$

where Y_t is output, A_t is total factor productivity (TFP), N_t is population (equivalently, labor supply), and $\alpha \in (0, 1)$ captures diminishing returns to labor given fixed factors such as land.

TFP evolves according to

$$A_{t+1} = A_t \left(1 + g(N_t; \theta)\right), \tag{2}$$

where the productivity growth function is

$$g(N_t; \theta) = \bar{g} + \theta \ln N_t. \tag{3}$$

Here $\bar{g} > 0$ is baseline productivity growth and $\theta \in \mathbb{R}$ is the *scale effect parameter*. When $\theta > 0$, larger populations generate faster productivity growth through mechanisms such as increased idea production (Romer, 1990; Kremer, 1993). When $\theta < 0$, larger populations slow productivity growth, perhaps through environmental degradation or resource depletion. When $\theta = 0$, productivity growth is independent of population.

Per capita consumption, which I take as the welfare-relevant variable, is

$$c_t = \frac{Y_t}{N_t} = A_t N_t^{\alpha - 1}. \tag{4}$$

Since $\alpha < 1$, larger populations reduce current per capita consumption holding TFP fixed. However, if $\theta > 0$, larger populations raise future TFP, creating an intertemporal tradeoff. The welfare implications of population growth depend on the relative magnitudes of these effects.

2.1.2 Households and Beliefs

The economy contains a unit mass of dynastic households indexed by $i \in [0, 1]$. Each household i holds a belief θ_i about the true value of the scale effect parameter θ. Beliefs are heterogeneous across households.

Assumption 1 (Belief Types)**.** *There exist two types of households. A fraction $\phi_t \in [0, 1]$ are* Romerians *with belief $\theta_R > 0$. The remaining fraction $1 - \phi_t$ are* Malthusians *with belief $\theta_M < 0$.*

I denote the true value of the scale effect parameter by $\theta^* \in \mathbb{R}$. A "Romerian world" has $\theta^* > 0$; a "Malthusian world" has $\theta^* < 0$. Households do not know θ^* with certainty but hold prior beliefs and update using the Bayes' Rule.

The distribution of beliefs, characterized by ϕ_t, is common knowledge in each period, but individual household types are private information. Households observe aggregate variables (Y_t, A_t, N_t) and the history thereof, but do not observe other households' beliefs or fertility choices directly.

2.1.3 Fertility Choice

Adult households choose consumption c_{it} and fertility $n_{it} \geq 0$ (children per adult). The household faces the budget constraint

$$c_{it} + \psi(n_{it}) = w_t, \tag{5}$$

where $w_t = \alpha A_t N_t^{\alpha-1}$ is the wage (marginal product of labor) and $\psi : \mathbb{R}_+ \to \mathbb{R}_+$ is the cost of raising children.

Assumption 2 (Child-Rearing Cost)**.** *The child-rearing cost function satisfies $\psi(0) = 0$, $\psi'(n) > 0$, $\psi''(n) > 0$ for all $n \geq 0$, and $\lim_{n\to\infty} \psi'(n) = \infty$.*

For analytical tractability, I use the quadratic specification $\psi(n) = \frac{\gamma}{2}n^2$ with $\gamma > 0$ in most of the analysis.

Households have dynastic preferences. Following Barro and Becker (1989), I assume the household derives utility from own consumption and from the welfare of descendants:

$$U_{it} = u(c_{it}) + \beta n_{it} \mathbb{E}_{it} \left[V_{i,t+1} \right], \tag{6}$$

where $u(\cdot)$ is a strictly concave period utility function (I use $u(c) = \ln c$), $\beta \in (0, 1)$ is the discount factor, and $V_{i,t+1}$ is the continuation value for the dynasty starting next period. The term n_{it} multiplying the continuation value captures the idea that the parent's utility from future generations scales with the number of children; this is the Barro-Becker formulation. Throughout, I assume that the equilibrium fertility levels satisfy $\beta n_j^* < 1$ so that the dynastic value function is well defined.

The expectation $\mathbb{E}_{it}[\cdot]$ is taken under household i's subjective beliefs about the parameter θ and about the actions of other households.

2.1.4 Population Dynamics

Aggregate population evolves according to

$$N_{t+1} = \bar{n}_t \cdot N_t, \tag{7}$$

where $\bar{n}_t$ is average fertility:

$$\bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}. \tag{8}$$

Here $n_{R,t}$ and $n_{M,t}$ denote the fertility choices of Romerians and Malthusians, respectively.

2.2 Belief Formation and Intergenerational Transmission

Beliefs evolve across generations through two channels: cultural transmission from parents and learning from observed economic outcomes.

2.2.1 Cultural Transmission

Children are born to households and initially adopt their parents' beliefs with probability $\lambda \in [0, 1]$. With probability $1 - \lambda$, they form beliefs independently through learning.

Assumption 3 (Cultural Transmission)**.** *A child born to a type-j household adopts type j with probability λ and forms beliefs through learning with probability $1 - \lambda$.*

When $\lambda = 1$, beliefs are perfectly persistent across generations. When $\lambda = 0$, beliefs are formed entirely through learning.

2.2.2 Bayesian Learning

Children who do not inherit parental beliefs form beliefs by applying Bayes' rule to observed data. Let π_t denote the common prior probability that the world is Romerian (i.e., $\theta^* = \theta_R$)

before observing generation-t outcomes. I assume that children who learn observe realized productivity growth:

$$s_t = g(N_t; \theta^*) + \varepsilon_t = \bar{g} + \theta^* \ln N_t + \varepsilon_t, \tag{9}$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is an i.i.d. noise term representing measurement error or transitory productivity shocks that obscure the true relationship.

Given signal s_t and population N_t, the likelihood of observing s_t under belief θ is

$$\ell(s_t|\theta) = \frac{1}{\sqrt{2\pi}\sigma_\varepsilon} \exp\left(-\frac{(s_t - \bar{g} - \theta \ln N_t)^2}{2\sigma_\varepsilon^2}\right). \tag{10}$$

Bayes' rule gives the posterior:

$$\pi_{t+1} = \frac{\pi_t \cdot \ell(s_t|\theta_R)}{\pi_t \cdot \ell(s_t|\theta_R) + (1 - \pi_t) \cdot \ell(s_t|\theta_M)}. \tag{11}$$

Children who learn adopt Romerian beliefs with probability π_{t+1} and Malthusian beliefs with probability $1 - \pi_{t+1}$.

2.2.3 Evolution of Belief Distribution

Combining cultural transmission and learning, the Romerian share evolves as follows. The fraction of next-generation adults who are Romerian children of Romerian parents is $\phi_t^{birth} \cdot \lambda$, where

$$\phi_t^{birth} = \frac{\phi_t n_{R,t}}{\bar{n}_t} \tag{12}$$

is the fraction of children born to Romerians. The fraction who are learning children of any parent who learn Romerian is $(1 - \lambda) \cdot \pi_{t+1}$. Summing:

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda)\pi_{t+1}. \tag{13}$$

This law of motion shows that the Romerian share is pushed upward by differential fertility (through ϕ_t^{birth}, which exceeds ϕ_t when $n_R > n_M$) and by learning (through π_{t+1}, which moves toward 1 when evidence favors Romerians).

2.3 Equilibrium

I now define equilibrium in this economy.

Definition 1 (Equilibrium)**.** *An equilibrium consists of:*

(i) Fertility policy functions $n_R(A_t, N_t, \phi_t)$ and $n_M(A_t, N_t, \phi_t)$ for each type;

(ii) Aggregate fertility $\bar{n}(A_t, N_t, \phi_t) = \phi_t n_R + (1 - \phi_t) n_M$*;*

(iii) Laws of motion for $(A_{t+1}, N_{t+1}, \phi_{t+1})$ *given by* (2)*,* (7)*, and* (13)*;*

such that:

(a) Each household's fertility choice maximizes expected dynastic utility (6) *subject to* (5)*, given beliefs* θ_i *and given the aggregate fertility function* $\bar{n}(\cdot)$*;*

(b) Expectations about aggregate fertility are consistent with the equilibrium aggregate fertility function;

(c) The actual laws of motion use the true parameter θ^**.*

Two features of this equilibrium concept merit emphasis. First, households optimize under their subjective beliefs but the actual economy evolves according to the true parameter θ^*. This allows for systematic forecast errors. Second, households must form expectations about aggregate fertility, which depends on other households' beliefs and actions. I impose rational expectations about others' behavior: each household correctly understands the mapping from the state to aggregate fertility, even though households disagree about structural parameters.

I emphasize that this notion of equilibrium describes a *decentralized* allocation in which individual households take aggregate fertility as given and do not internalize the externality that population growth imposes on TFP through (2). Even when all households share the correct belief θ^*, the resulting allocation is therefore not in general the social planner's optimum—a point that becomes important when interpreting the welfare benchmark in Section 7.

2.4 Household Optimization

I now characterize the household's optimal fertility choice. Let $V_j(A, N, \phi)$ denote the value function for a type-j household. The Bellman equation is:

$$V_j(A, N, \phi) = \max_{n \geq 0} \left\{ \ln(w - \psi(n)) + \beta n \cdot \mathbb{E}_j \left[V_j(A', N', \phi') \right] \right\}, \tag{14}$$

where the expectation is taken under belief θ_j and where (A', N', ϕ') evolve according to the laws of motion.

I conjecture a value function of the form:

$$V_j(A, N, \phi) = v_j(\phi) + \xi_{A,j} \ln A + \xi_{N,j} \ln N, \tag{15}$$

where $\xi_{A,j}$ and $\xi_{N,j}$ are type-specific constants (because beliefs and the optimal own fertility differ across types) and $v_j(\phi)$ captures type-specific continuation values. As is standard, (15) is to be interpreted as a log-linear approximation around a balanced-growth reference point; the residual non-log-linear contribution from $-\psi(n)/w$ is absorbed into the type-specific function $v_j(\phi)$.

Under this conjecture, the first-order condition for fertility is:

$$\frac{\psi'(n_j)}{w - \psi(n_j)} = \beta \mathbb{E}_j \left[V_j(A', N', \phi') \right]. \tag{16}$$

With the quadratic cost $\psi(n) = \frac{\gamma}{2} n^2$, this becomes:

$$\frac{\gamma n_j}{w - \frac{\gamma}{2} n_j^2} = \beta \mathbb{E}_j \left[V_j(A', N', \phi') \right]. \tag{17}$$

The following lemma characterizes equilibrium fertility.

Lemma 1 (Equilibrium Fertility)**.** *Under Assumptions 1–3 and the value function conjecture (15), equilibrium fertility for type $j \in \{R, M\}$ satisfies:*

$$n_j^* = n^* \cdot h(\theta_j; \bar{n}), \tag{18}$$

where n^* is a baseline fertility level, and $h(\theta; \bar{n})$ is increasing in θ with $h(\bar{\theta}; \bar{n}) = 1$ for some reference belief $\bar{\theta}$. Moreover, $n_R^* > n_M^*$ for all equilibrium values of $\bar{n}$.

The proof is in Appendix A. The key economic insight is that households who believe population enhances productivity (Romerians) perceive a higher return to children and hence choose higher fertility. This prediction—that Malthusian beliefs lead to lower fertility—is testable. I examine empirical support for this prediction in Section 5.

3 Strategic Complementarity and Substitutability

I now analyze strategic interaction in fertility choices. The key question is: how does a household's optimal fertility respond to expectations about aggregate fertility? I show that, under empirically plausible parameter conditions, the answer differs by type: Romerians exhibit strategic complementarity while Malthusians exhibit strategic substitutability. This asymmetry has important implications for equilibrium properties.

3.1 The Fertility Game

Consider the fertility choice as a game among households. Each household i chooses n_i to maximize expected utility. Aggregate fertility is $\bar{n} = \int n_i \, di$. The household's payoff depends on $(n_i, \bar{n})$ through the following channel: higher $\bar{n}$ raises $N' = \bar{n} \cdot N$, which affects A'' through (2) (and hence subsequent productivity) and dilutes future per capita consumption.

To formalize, let $U_j(n_j, \bar{n})$ denote the expected utility of a type-j household choosing fertility n_j when aggregate fertility is $\bar{n}$. This expected utility accounts for future consumption, which depends on future TFP and population under the household's subjective belief θ_j.

Definition 2 (Strategic Complementarity/Substitutability)**.** *Household type j exhibits* strategic complementarity *if* $\frac{\partial^2 U_j}{\partial n_j \partial \bar{n}} > 0$ *and* strategic substitutability *if* $\frac{\partial^2 U_j}{\partial n_j \partial \bar{n}} < 0$.

Strategic complementarity means that the marginal return to fertility is increasing in aggregate fertility. Strategic substitutability means the opposite.

3.2 Analysis Using Global Games Techniques

To analyze strategic interaction rigorously, I adapt the global games framework of Carlsson and Van Damme (1993) and Morris and Shin (1998). The standard global games setup features a continuum of players with private signals about a fundamental, facing a binary action choice in a game with strategic complementarities. I extend this to allow for continuous actions (fertility) and for heterogeneous strategic interaction (complementarity for Romerians, substitutability for Malthusians).

3.2.1 Setup

Suppose the economy is in a state where current TFP A and population N are known, and the Romerian share is ϕ. Each household knows its own type (and hence belief θ_j) but faces uncertainty about the types of others and hence about aggregate fertility $\bar{n}$.

Let $n_j(\bar{n}^e)$ denote the fertility choice of type j when the expected aggregate fertility is $\bar{n}^e$. The actual aggregate fertility is then:

$$\bar{n} = \phi \cdot n_R(\bar{n}^e) + (1 - \phi) \cdot n_M(\bar{n}^e). \tag{19}$$

A rational expectations equilibrium requires $\bar{n}^e = \bar{n}$: expected aggregate fertility equals actual aggregate fertility.

3.2.2 Best Response Functions

To derive best response functions, I compute how each type's optimal fertility depends on expected aggregate fertility. Recall that current-period TFP growth depends on *current* population N_t, not on current aggregate fertility $\bar{n}_t$. Aggregate fertility today, $\bar{n}_t$, affects future variables only through $N_{t+1} = \bar{n}_t N_t$, which then influences A_{t+1} (under non-zero scale effects) and per capita consumption.

Using the value function conjecture, the continuation value summarizes both effects. Under belief θ_j, the expected continuation value $\mathbb{E}_j[V_j(A', N', \phi')]$ is increasing in $\bar{n}^e$ precisely when the discounted scale-effect benefit outweighs the per capita dilution penalty. Standard recursive calculation (carried out in the Appendix) shows that the marginal effect has the same sign as

$$(\alpha - 1)(1 - \beta n_j) + \beta n_j \theta_j. \tag{20}$$

This sign is unambiguously negative for Malthusians (since $\theta_j < 0$ and $\alpha - 1 < 0$, both summands are negative). For Romerians, it is positive when θ_R is sufficiently large, specifically when

$$\theta_R > \frac{(1 - \alpha)(1 - \beta n_R)}{\beta n_R}. \tag{21}$$

For empirically plausible values—$\alpha \approx 0.7$, $\beta \approx 0.95$, $n_R \approx 1$—the right-hand side is on the order of 0.016, which is comfortably below the calibrated $\theta_R \approx 0.1$ used later. Throughout the rest of the analysis, I maintain the standing assumption that (21) holds.

Proposition 1 (Strategic Interaction)**.** *Consider the fertility game under Assumptions 1–3. Suppose condition (21) holds. For type $j \in \{R, M\}$, let $n_j^*(\bar{n}^e)$ denote the optimal fertility given expected aggregate fertility $\bar{n}^e$. Then:*

(i) Romerians exhibit strategic complementarity: $\frac{dn_R^*}{d\bar{n}^e} > 0$.

(ii) Malthusians exhibit strategic substitutability: $\frac{dn_M^*}{d\bar{n}^e} < 0$.

The proof is in Appendix A. The intuition is as follows. When a Romerian expects higher aggregate fertility, they expect higher future population, which under Romerian beliefs translates to faster productivity growth in subsequent generations. Provided this benefit dominates the dilution of per capita consumption (condition (21)), the continuation value of each child rises, increasing the marginal benefit of fertility. The Romerian responds by having more children. When a Malthusian expects higher aggregate fertility, both forces—slower future productivity growth and dilution of per capita consumption—work in the same direction, lowering the continuation value and reducing fertility.

3.3 Equilibrium Characterization

Proposition 1 implies that the aggregate best response function—the mapping from expected aggregate fertility to actual aggregate fertility—depends on the composition of the population.

Corollary 1 (Slope of Aggregate Best Response)**.** *Under the conditions of Proposition 1, the aggregate best response function* $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1-\phi)n_M^*(\bar{n}^e)$ *has slope:*

$$\frac{dBR}{d\bar{n}^e} = \phi\frac{dn_R^*}{d\bar{n}^e} + (1-\phi)\frac{dn_M^*}{d\bar{n}^e}. \tag{22}$$

This slope is:

(i) Positive when ϕ is sufficiently large (Romerian dominance);

(ii) Negative when ϕ is sufficiently small (Malthusian dominance);

(iii) Zero at a critical ϕ^* where the two effects exactly offset.

The critical share ϕ^* satisfies:

$$\phi^* = \frac{|dn_M^*/d\bar{n}^e|}{|dn_R^*/d\bar{n}^e| + |dn_M^*/d\bar{n}^e|}. \tag{23}$$

When $\phi > \phi^*$, the aggregate best response function slopes upward, creating the potential for multiple equilibria. A slight increase in expected fertility raises actual fertility, which can be self-fulfilling. When $\phi < \phi^*$, the aggregate best response slopes downward, ensuring a unique equilibrium: any deviation from equilibrium is self-correcting.

Proposition 2 (Equilibrium Multiplicity)**.** *Under the conditions of Proposition 1:*

(i) If $\phi < \phi^*$, there exists a unique equilibrium.

(ii) If $\phi > \phi^*$ and the slope of $BR(\bar{n}^e)$ exceeds 1 at some point, there exist multiple equilibria.

In the multiple-equilibria region, equilibria can be ranked by aggregate fertility: there exists a low-fertility equilibrium, a high-fertility equilibrium, and potentially unstable intermediate equilibria.

The proof follows standard arguments from the global games literature and is provided in Appendix A. The economic implication is that societies dominated by Romerians may be subject to self-fulfilling fluctuations in fertility, whereas societies dominated by Malthusians have more stable fertility dynamics.

3.4 Uniqueness under Incomplete Information

A key insight from the global games literature is that introducing small amounts of private information can select a unique equilibrium even when multiple equilibria exist under common knowledge. I now show how this logic applies to the fertility game.

Suppose that each household observes a private signal about the Romerian share:

$$\phi_i = \phi + \eta_i, \tag{24}$$

where $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$ is i.i.d. across households. Households must form expectations about ϕ (and hence about aggregate fertility) based on their private signal and prior beliefs. To make contact with the standard global games machinery, in the multiple-equilibria region of Proposition 2 I focus on selection between the low-fertility and high-fertility equilibria of the continuous-action game; I refer to these locally as the "low" and "high" actions.

Proposition 3 (Equilibrium Selection)**.** *Consider the parameter region in which Proposition 2(ii) holds, so that the complete-information game admits a low-fertility and a high-fertility equilibrium. As signal noise $\sigma_\eta \to 0$, there exists a unique equilibrium of the incomplete-information game in which each type uses a threshold strategy on its private signal:*

(i) Romerians coordinate on the high-fertility equilibrium n_R^H when $\phi_i > \phi^{**}$ and on the low-fertility equilibrium n_R^L when $\phi_i < \phi^{**}$;

(ii) Malthusians choose their (substitute) low-fertility response n_M^L when $\phi_i > \phi^{**}$ and their high-fertility response n_M^H when $\phi_i < \phi^{**}$;

where the common threshold ϕ^{**} depends on model parameters.

The proof adapts the arguments of Morris and Shin (1998) and is provided in Appendix A. The key insight is that private information breaks the common-knowledge structure that supports multiple equilibria. Each household is uncertain about whether other households are above or below the threshold, and this uncertainty pins down a unique equilibrium.

The threshold ϕ^{**} depends on the relative strength of strategic complementarity among Romerians and strategic substitutability among Malthusians. When Romerian strategic complementarity is strong, the threshold is lower, meaning that Romerians coordinate on high fertility even when the Romerian share is moderate.

4 Long-Run Dynamics and the Role of Wrong Beliefs

I now analyze the long-run dynamics of beliefs and population. The key insight is that dynamics depend critically on whether the true world is Romerian ($\theta^* > 0$) or Malthusian ($\theta^* < 0$). In the former case, incorrect beliefs cause only transitory inefficiency. In the latter case, incorrect beliefs can generate persistent cycles.

4.1 Two Forces: Demographic Selection and Learning

The Romerian share ϕ_t evolves according to (13):

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda)\pi_{t+1}. \tag{25}$$

Two forces drive this evolution.

Demographic selection. The term $\phi_t^{birth} = \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t) n_M}$ captures the effect of differential fertility. Since $n_R > n_M$ (Romerians have more children), we have $\phi_t^{birth} > \phi_t$ for any $\phi_t \in (0, 1)$. Demographic selection pushes ϕ_{t+1} upward regardless of which belief is correct.

Bayesian learning. The term π_{t+1} captures the effect of evidence. The direction depends on the true θ^*:

- If $\theta^* > 0$ (Romerian world): high population tends to generate high productivity growth, which is evidence for the Romerian model. π_{t+1} tends to exceed π_t in expectation.
- If $\theta^* < 0$ (Malthusian world): high population tends to generate low productivity growth, which is evidence for the Malthusian model. π_{t+1} tends to fall below π_t in expectation.

4.2 Dynamics in a Romerian World

Theorem 1 (Convergence in Romerian Worlds)**.** *Suppose $\theta^* > 0$ and $\ln N_t \neq 0$ along the equilibrium path. Then, starting from any initial condition (ϕ_0, N_0, A_0) with $\phi_0 \in (0, 1)$, the economy converges almost surely to a steady state with:*

(i) $\phi_t \to 1$*: Romerians dominate the population.*

(ii) $\pi_t \to 1$*: posterior probability of Romerian world converges to certainty.*

(iii) Aggregate fertility converges to n_R^**.*

(iv) Population grows at rate $n_R^* - 1$ in the long run.

The proof is in Appendix A. The intuition is that both demographic selection and learning favor Romerians when the world is truly Romerian. Demographic selection favors them because they have more children. Learning favors them because observed outcomes confirm their beliefs.

Welfare implications. Malthusians in a Romerian world impose a social cost by suppressing aggregate fertility below the efficient level. The efficient fertility level maximizes the sum of dynastic utilities, taking into account the positive externality of population on future productivity. Since Malthusians undervalue this externality, they underinvest in children.

However, this cost is transitory. As Malthusians shrink in population share and as children of Malthusians convert to Romerianism through learning, aggregate fertility rises toward the efficient level. The welfare loss from initial belief heterogeneity vanishes in the long run.

4.3 Dynamics in a Malthusian World

The dynamics are starkly different when $\theta^* < 0$.

Theorem 2 (Cycles in Malthusian Worlds)**.** *Suppose $\theta^* < 0$. Then, for intermediate values of the learning responsiveness (parameterized by σ_ε^{-2}), the linearized dynamics around an interior steady state admit complex eigenvalues with modulus exceeding one, generating local oscillations. In the nonlinear system, these oscillations are bounded and produce endogenous cycles in which:*

(i) The Romerian share ϕ_t oscillates, never converging to 0 or 1.

(ii) Population N_t oscillates, with booms and busts.

(iii) Per capita consumption c_t oscillates countercyclically with ϕ_t.

The cycle period depends on the transmission parameter λ and the learning responsiveness.

The proof is in Appendix A. The key mechanism is the conflict between demographic selection and learning.

Cycle dynamics. Starting from an interior ϕ_0:

Expansion phase: Suppose population is moderate and welfare is acceptable. Both models fit the data reasonably well, so learning is weak. Demographic selection dominates: $\phi_{t+1} > \phi_t$ because Romerians have more children. Rising ϕ raises aggregate fertility, causing population to grow.

Crisis phase: As population grows, productivity growth slows (since $\theta^* < 0$). Per capita consumption falls. The Malthusian model fits the data much better than the Romerian model: high population coincided with poor outcomes. Young generations update sharply toward Malthusian beliefs: π_{t+1} falls. If learning is sufficiently responsive, this overwhelms demographic selection, and $\phi_{t+1} < \phi_t$.

Recovery phase: Falling ϕ reduces aggregate fertility. Population growth slows or reverses. Per capita consumption stabilizes and begins recovering. With moderate population, both models again fit the data similarly, weakening learning. Demographic selection again dominates, and ϕ begins rising, starting the next cycle.

Conditions for cycles. Not all Malthusian worlds exhibit cycles. If learning is very weak (high σ_ε), demographic selection always dominates, and ϕ increases monotonically toward 1. The economy converges to a steady state with Romerian beliefs, even though these beliefs are incorrect. Population is inefficiently high, and welfare is inefficiently low, but there are no cycles.

If learning is very strong (low σ_ε), young generations quickly identify the true model, and Malthusian beliefs spread rapidly. The interior steady state is destabilized in favor of a corner steady state with Malthusian dominance, low fertility, and stable population.

Cycles occur for intermediate learning responsiveness, where demographic selection and learning are both operative but neither dominates and the interior steady state is locally an unstable focus.

Proposition 4 (Cycle Existence Conditions)**.** *Define $\mathcal{L} = \sigma_\varepsilon^{-2}|\theta_R - \theta_M|$ as an index of learning responsiveness and $\mathcal{D} = \lambda(n_R - n_M)/\bar{n}$ as an index of demographic selection strength. The linearized economy exhibits oscillatory instability around an interior steady state (the precondition for the limit cycle described in Theorem 2) if and only if:*

$$\underline{\mathcal{L}}(\mathcal{D}) < \mathcal{L} < \overline{\mathcal{L}}(\mathcal{D}), \tag{26}$$

where the threshold functions $\underline{\mathcal{L}}$ and $\overline{\mathcal{L}}$ are characterized in Appendix A.

The proof is in Appendix A.

4.4 The Asymmetry: Why Wrong Beliefs Matter Differently

The analysis reveals a fundamental asymmetry between the two scenarios.

In Romerian worlds, both forces align toward correct beliefs. Demographic selection favors optimists, and optimists happen to be right. Learning favors the correct model, which is optimistic. Incorrect (Malthusian) beliefs are self-correcting.

In Malthusian worlds, the forces conflict. Demographic selection favors optimists, but optimists are wrong. Learning favors the correct model, which is pessimistic. The result is either persistent incorrect beliefs (if demographic selection dominates) or cycles (if neither force dominates).

This asymmetry has a simple but profound implication: *human societies are more vulnerable to excessive optimism than excessive pessimism.* Optimists reproduce more, regardless of whether their optimism is warranted. Pessimism is only reinforced through learning, which requires observing bad outcomes—i.e., crises. The system is biased toward optimism.

Remark 1. *The asymmetry helps explain why Malthusian crises recurred throughout pre-industrial history despite ample evidence that population growth beyond carrying capacity leads to disaster. Each generation's optimists outbred its pessimists, regenerating the conditions for the next crisis. Only the crisis itself, by providing unmistakable evidence against the Romerian view, temporarily restored Malthusian dominance.*

5 Empirical Evidence: Beliefs and Fertility

A central prediction of the model is that Malthusian beliefs reduce fertility. Lemma 1 establishes that households believing population diminishes productivity ($\theta_M < 0$) choose lower fertility than households believing population enhances productivity ($\theta_R > 0$). In this section, I examine empirical support for this prediction using survey data on beliefs and fertility.

5.1 Data

I use data from the General Social Survey (GSS), a nationally representative survey of U.S. adults conducted by NORC at the University of Chicago. The GSS has been administered since 1972 and includes questions on a wide range of social attitudes and demographic characteristics.

The key belief variable is drawn from the question: "The earth cannot continue to support population growth at its present rate." Respondents indicate their agreement on a five-point scale ranging from "strongly agree" to "strongly disagree." I classify respondents who agree or strongly agree as holding *Malthusian beliefs*, corresponding to the view that population growth is unsustainable. This question was asked in the 2000 and 2010 survey waves.

I examine two fertility measures. The first is stated preferences: respondents' answer to "What do you think is the ideal number of children for a family to have?" Responses range from zero to seven or more. The second is realized fertility: the actual number of children the respondent has had. Examining both measures allows me to test whether beliefs shape not only fertility intentions but also fertility outcomes.

I restrict the sample to respondents with valid responses to both the belief and fertility questions, as well as to demographic controls including age, education, and sex. The resulting sample contains 1,822 observations.

5.2 Descriptive Patterns

Table 1 presents summary statistics. The mean ideal number of children is 2.50, with substantial variation (standard deviation 0.87). The mean actual number of children is 1.81 (standard deviation 1.64), reflecting that many respondents have not yet completed their fertility. Approximately 55 percent of respondents hold Malthusian beliefs (agree or strongly agree that Earth cannot support continued population growth). The sample is 56 percent female, with mean age of 47 and mean education of 13.3 years.

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from General Social Survey, 2000 and 2010 waves. Malthusian belief (binary) equals one if respondent agrees or strongly agrees that Earth cannot continue to support population growth. Malthusian belief (1–5 scale) is reverse-coded so that higher values indicate stronger Malthusian beliefs.

Figure 1 displays mean ideal and actual fertility by the intensity of Malthusian beliefs. The pattern is striking: there is a monotonic negative relationship between Malthusian belief strength and both fertility measures. Respondents who strongly disagree that Earth cannot support population growth (the least Malthusian) report an average ideal of 2.98 children and have 1.76 children, while those who strongly agree (the most Malthusian) report an average ideal of 2.24 children and have 1.52 children. The gradient is present for both stated preferences and realized outcomes, though the relationship is stronger for ideal fertility.

A potential concern is that this relationship is driven by education: more educated individuals may both hold more Malthusian beliefs (through exposure to environmental

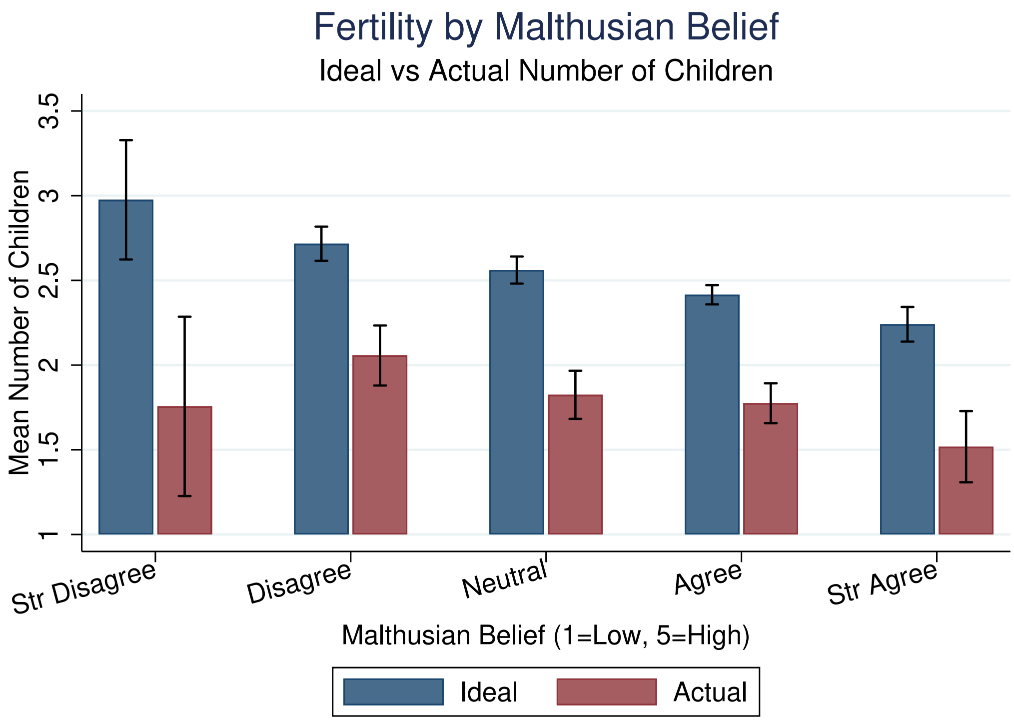

Figure 1. Fertility by Malthusian Belief Intensity

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal (navy) and actual (maroon) number of children for each level of the Malthusian belief scale, where 1 = "strongly disagree" and 5 = "strongly agree" that Earth cannot continue to support population growth. Error bars indicate 95% confidence intervals.

discourse) and desire fewer children (through higher opportunity costs). Figure 2 addresses this concern by displaying mean ideal fertility separately by education group and belief type. Within every education category—less than high school, high school graduate, some college, and college or more—Malthusians report lower ideal fertility than non-Malthusians. The gap ranges from 0.20 children among college graduates to 0.45 children among those with less than high school education. This pattern suggests that the belief-fertility relationship is not simply a proxy for education.

5.3 Regression Analysis

To assess the robustness of these patterns, I estimate linear regressions of both ideal and actual fertility on Malthusian beliefs with demographic controls. Table 2 presents the results.

Columns (1) and (2) report results for ideal fertility. Malthusians report ideal fertility 0.27 children lower than non-Malthusians (column 1), statistically significant at the 1 percent level. This represents approximately 11 percent of the sample mean. Using the continuous

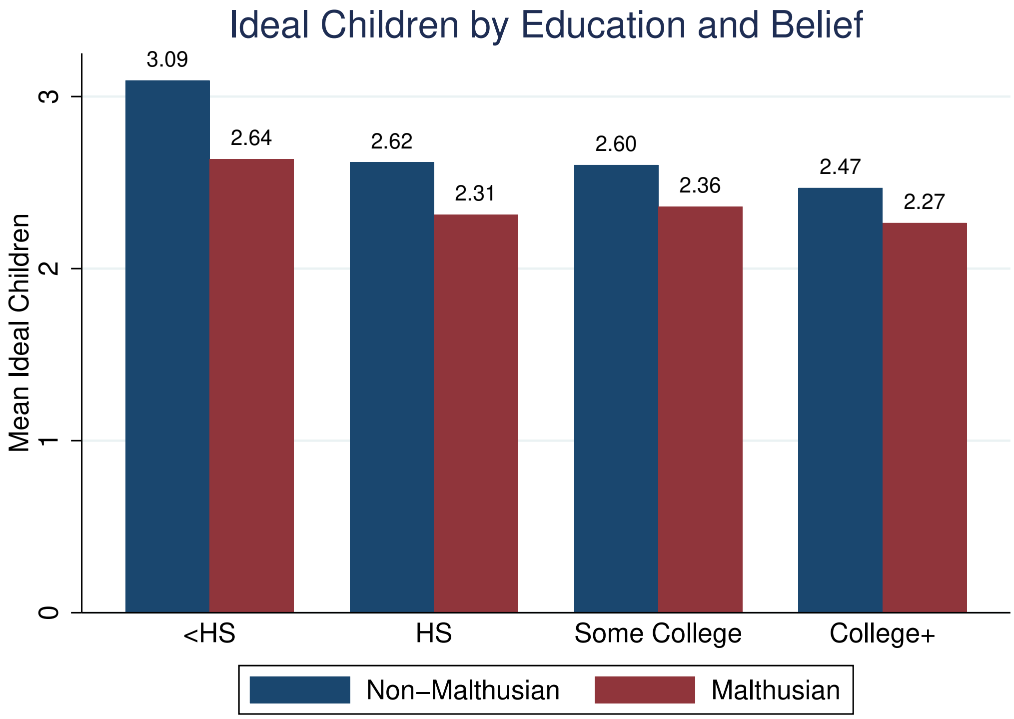

Figure 2. Ideal Number of Children by Education and Malthusian Belief

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal number of children for non-Malthusians (navy) and Malthusians (maroon) within each education category. Malthusian is defined as agreeing or strongly agreeing that Earth cannot continue to support population growth. Numbers above bars indicate group means.

belief scale (column 2), each one-unit increase in Malthusian belief intensity is associated with 0.15 fewer ideal children.

Columns (3) and (4) report results for actual fertility. The coefficient on Malthusian beliefs is −0.29 (column 3), implying that Malthusians have approximately 0.29 fewer children than non-Malthusians—a difference of 16 percent relative to the sample mean of 1.81. The effect using the continuous scale is −0.15 per unit (column 4). Notably, the coefficient on actual fertility is *larger* in absolute value than the coefficient on ideal fertility, suggesting that beliefs translate into behavior with at least as much force as they affect stated preferences.

The higher R^2 for actual fertility (0.26 versus 0.06) reflects the strong relationship between age and realized fertility: older respondents have had more time to accumulate children. The coefficient on age for actual fertility is positive (0.11 per year), while for ideal fertility it is slightly negative (−0.01), consistent with younger cohorts expressing somewhat higher fertility ideals.

Table 2. Effect of Malthusian Beliefs on Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	-0.268^{***}		-0.292^{***}	
	(0.041)		(0.066)	
Malthusian (continuous)		-0.153^{***}		-0.146^{***}
		(0.021)		(0.033)
Age	-0.014^{**}	-0.015^{**}	0.112^{***}	0.111^{***}
	(0.006)	(0.006)	(0.010)	(0.010)
Female	0.015	0.026	0.091	0.104
	(0.040)	(0.040)	(0.068)	(0.068)
Education	-0.047^{***}	-0.046^{***}	-0.144^{***}	-0.143^{***}
	(0.007)	(0.007)	(0.013)	(0.013)
Year FE	Yes	Yes	Yes	Yes
Observations	1,822	1,822	1,822	1,822
R^2	0.060	0.069	0.262	0.262

Notes: All models include age squared and year fixed effects. Malthusian is a binary indicator for agreeing or strongly agreeing that Earth cannot continue to support population growth. Malthusian (continuous) is a 1–5 scale where higher values indicate stronger Malthusian beliefs. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.4 Completed Fertility

A concern with analyzing actual fertility is that younger respondents may not have completed their childbearing. I address this by restricting the sample to respondents with likely completed fertility: women aged 45 and older, and men aged 50 and older. Table 3 presents results for this subsample.

Among those with completed fertility (column 2), Malthusians have 0.26 fewer children than non-Malthusians, significant at the 5 percent level. The coefficient is similar in magnitude to the full-sample estimate, confirming that the relationship holds for realized lifetime fertility, not just current fertility among a mix of ages. The relationship is also present among those who have not yet completed fertility (column 3), with a coefficient of -0.29, suggesting that belief-driven fertility differentials emerge early in the reproductive life course.

5.5 Heterogeneity

Table 4 examines whether the belief-fertility relationship varies across demographic groups. I estimate the full specification separately by education level and sex for both fertility mea-

Table 3. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	-0.292^{***}	-0.257^{**}	-0.288^{***}
	(0.066)	(0.109)	(0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

Notes: Dependent variable is actual number of children. All models control for age, age squared, sex, education, and year. Completed fertility defined as women age 45+ or men age 50+. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

sures.

Table 4. Heterogeneity in the Belief-Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	-0.318^{***}	-0.235^{***}	-0.209^{*}	-0.392^{***}
	(0.069)	(0.049)	(0.115)	(0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: Dependent variable is ideal (columns 1–2) or actual (columns 3–4) number of children. All models control for age, age squared, sex, and year. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The belief-fertility relationship is present across education groups for both outcome measures. For ideal fertility, the coefficient is somewhat larger for less educated respondents (-0.32) than for more educated respondents (-0.24). For actual fertility, the pattern reverses: the coefficient is larger for more educated respondents (-0.39) than for less educated respondents (-0.21). This suggests that while Malthusian beliefs affect fertility preferences across the education distribution, the translation of beliefs into behavior may be stronger among the more educated, perhaps because they face fewer constraints on achieving their fertility targets.

5.6 Interpretation and Limitations

The empirical findings are consistent with the model's central prediction: individuals who believe population growth is unsustainable (Malthusians) desire and have fewer children than those who do not hold such beliefs. The relationships are economically meaningful—gaps of 0.27 ideal children (11 percent) and 0.29 actual children (16 percent)—and statistically robust across specifications.

Two features of the results merit emphasis. First, the relationship holds for *actual* fertility, not just stated preferences. This addresses a common concern that surveys capture cheap talk rather than consequential behavior. The fact that Malthusians have fewer children, even among those with completed fertility, suggests that beliefs do translate into fertility outcomes.

Second, the coefficient on actual fertility is at least as large as the coefficient on ideal fertility. This is notable because one might expect beliefs to affect intentions more strongly than behavior, given the many constraints (partner preferences, income, fecundity) that intervene between preferences and outcomes. The finding that the behavioral effect is similar or larger suggests either that beliefs operate partly through channels beyond conscious intention, or that individuals with stronger beliefs are more committed to acting on them.

Several limitations warrant emphasis. The evidence remains correlational. The documented association could reflect reverse causality (individuals who have or want fewer children rationalize this by adopting Malthusian beliefs) or omitted variable bias (unobserved characteristics affecting both beliefs and fertility). The cross-sectional nature of the data precludes tracking how belief changes affect subsequent fertility decisions.

Additionally, the belief measure captures concern about global population sustainability, not specifically the scale-effect parameter θ in the model. Respondents may agree that Earth cannot support population growth for reasons unrelated to productivity—for instance, concerns about biodiversity, quality of life, or resource exhaustion—that do not map directly onto the Romerian-Malthusian dichotomy.

Despite these limitations, the evidence provides suggestive support for the belief channel emphasized in the model. The robust correlations between Malthusian beliefs and both ideal and actual fertility, documented across demographic groups and specifications, are what the model predicts. More definitive causal identification remains an important direction for future research.

6 Demographic History

The model provides a lens for interpreting broad patterns in demographic history. I discuss three eras: the Malthusian period before industrialization, the demographic transition accompanying industrialization, and the modern fertility decline.

6.1 The Malthusian Era

For most of human history, roughly from the Neolithic Revolution (circa 10,000 BCE) to the Industrial Revolution (circa 1800 CE), economic historians characterize the world as Malthusian. Real wages fluctuated around subsistence levels, population grew slowly on average but with substantial fluctuations, and periods of demographic expansion were followed by crises that reduced population through famine, disease, and war (Clark, 2007; Voigtländer and Voth, 2013).

The model interprets this era as a Malthusian world ($\theta^* < 0$) with intermediate learning responsiveness, generating cycles of the type described in Theorem 2. During periods of relative prosperity, Romerian-like optimism spread through demographic selection. Population grew. Eventually, population exceeded carrying capacity, living standards fell, and Malthusian beliefs were reinforced by crisis. Population stabilized or declined, recovery began, and the cycle restarted.

The Black Death of 1347–1351 provides a particularly clear example. Europe's population had grown substantially in the preceding centuries, straining resources. The plague killed roughly one-third of the population. Real wages rose sharply in the aftermath and remained elevated for over a century. The model interprets this as a crisis phase that both directly reduced population and reinforced Malthusian beliefs, leading to a sustained period of lower fertility.

6.2 The Industrial Revolution and Demographic Transition

The Industrial Revolution marked a regime change from $\theta^* < 0$ to $\theta^* > 0$. Sustained technological progress, enabled by the accumulation of scientific knowledge and the development of institutions supporting innovation, broke the Malthusian link between population and living standards. Larger populations no longer reduced per capita income; instead, they contributed to faster innovation and growth.

The demographic transition—the shift from high fertility and high mortality to low fertility and low mortality—unfolded over roughly two centuries in early-industrializing countries. The model suggests several mechanisms through which belief updating and demographic selection interacted during this transition.

Initially, beliefs inherited from the Malthusian era may have suppressed fertility below the new optimum. Households with Malthusian priors underestimated the returns to children in the new Romerian world. However, two forces pushed beliefs toward Romerianism. First, observed outcomes increasingly favored the Romerian model: population grew while living standards rose, contradicting Malthusian predictions. Second, Romerians outbred

Malthusians, increasing the Romerian share.

The nineteenth-century population explosion, in which European population roughly tripled, can be partially attributed to this belief transition. As Romerian beliefs spread, aggregate fertility rose, contributing to rapid population growth. The model suggests that some of this growth may have overshot the efficient level as Romerian optimism became excessive, but this is difficult to distinguish from efficient adjustment to a new technological regime.

6.3 The Modern Fertility Decline

Fertility rates in developed countries have fallen dramatically since the 1960s, reaching levels well below replacement in most of Europe and East Asia. This decline is puzzling from a simple Romerian perspective: if larger populations are beneficial, why are the most prosperous societies choosing the fewest children?

The model suggests several interpretations, none mutually exclusive.

Revival of Malthusian beliefs. The environmental movement, which emerged prominently in the 1960s and 1970s, can be understood as a revival of Malthusian thinking. Concerns about pollution, resource depletion, climate change, and ecological limits represent beliefs that population growth is harmful. The empirical evidence in Section 5 documents that such beliefs are prevalent—over half of GSS respondents agree that Earth cannot support continued population growth—and are associated with substantially lower fertility preferences and outcomes. If a significant fraction of the population has adopted such beliefs, our model predicts lower fertility through two channels: the direct effect (Malthusians choosing fewer children) and strategic substitutability (Malthusians reducing fertility further when they perceive aggregate fertility as excessive).

Actual shift to Malthusian technology. An alternative interpretation is that the true θ^* has recently shifted from positive to negative. Automation and artificial intelligence may have reduced the marginal contribution of human labor to economic output. Environmental constraints, particularly climate change, may have imposed genuine costs on population growth. If the world has become Malthusian again, low fertility may be an appropriate response rather than a mistake.

Quantity-quality substitution. The model abstracts from human capital, but in reality, parents face a quantity-quality tradeoff. Rising returns to education may have induced parents to invest more in each child's human capital at the expense of having more children. This is conceptually distinct from Malthusianism but generates similar fertility outcomes.

Distinguishing among these interpretations is empirically challenging but important for policy. If the first interpretation is correct, fertility may be inefficiently low, and policies that correct misperceptions or subsidize childbearing may be warranted. If the second interpretation is correct, low fertility is efficient. If the third interpretation is correct, the efficiency of low fertility depends on whether private returns to human capital equal social returns.

6.4 Contemporary Divergence and Future Dynamics

Fertility today varies enormously across countries and groups. Total fertility rates range from below 1.0 in some East Asian countries to above 6.0 in parts of Sub-Saharan Africa. Within countries, fertility varies systematically with religion, education, and political orientation.

The model predicts that these differences are partially driven by belief heterogeneity and are self-reinforcing through demographic selection. Groups with Romerian beliefs (or with religious or cultural traditions that encourage high fertility for other reasons) will grow in population share over time. If cultural transmission is strong (λ high), these compositional shifts can substantially affect aggregate beliefs and fertility over generations.

This has implications for long-run population projections. Standard demographic projections typically assume convergence to low fertility as countries develop. The model suggests that convergence may be slower or incomplete if belief heterogeneity persists. Groups with high-fertility beliefs may maintain or increase their population share even as mean beliefs shift.

Whether this is efficient depends on the true value of θ^*. If the world remains Romerian, high-fertility groups are moving the population toward efficiency. If the world has become Malthusian, high-fertility groups are exacerbating the problem. The key empirical question is which technological regime currently prevails.

7 Policy: The Value of Public Information

I now analyze the welfare effects of public announcements about the scale-effect parameter θ. A key question is whether a social planner can improve upon the decentralized equilibrium by providing information, even when the planner herself faces uncertainty about the true parameter value.

7.1 Setup

Consider a benevolent social planner who observes a signal about the true parameter:

$$\hat{\theta} = \theta^* + \nu, \tag{27}$$

where $\nu \sim \mathcal{N}(0, \sigma_\nu^2)$ represents the planner's estimation error. The planner's signal precision is $\tau_\nu = \sigma_\nu^{-2}$. When $\tau_\nu \to \infty$, the planner knows θ^* perfectly; when $\tau_\nu \to 0$, the planner's signal is uninformative.

The planner can publicly announce a value θ^A. I consider two announcement policies: truthful announcement ($\theta^A = \hat{\theta}$) and no announcement (the status quo with heterogeneous private beliefs).

Upon hearing the announcement, households update their beliefs. I assume households treat the announcement as an additional signal and apply Bayes' rule. Specifically, after the announcement, a household with prior belief θ_j forms posterior belief:

$$\theta_j^{post} = \frac{\tau_j \theta_j + \tau_A \theta^A}{\tau_j + \tau_A}, \tag{28}$$

where τ_j is the precision of the household's prior and τ_A is the credibility (perceived precision) of the announcement.

Assumption 4 (Announcement Credibility)**.** *Households treat the public announcement as a signal with precision $\tau_A > 0$. When $\tau_A \to \infty$, households fully adopt the announced value; when $\tau_A \to 0$, households ignore the announcement.*

For analytical clarity, the headline results below focus on the case in which households fully adopt the announced value (τ_A large). This captures scenarios where the government has substantial credibility on long-run economic relationships and isolates the coordination channel that is the central focus of the paper. I discuss the more general case below.

7.2 Welfare under Heterogeneous Beliefs

In the decentralized equilibrium without announcement, welfare depends on the distribution of beliefs. Define aggregate welfare as the population-weighted sum of dynastic utilities:

$$W = \phi V_R(A, N, \phi) + (1 - \phi) V_M(A, N, \phi). \tag{29}$$

The inefficiency from belief heterogeneity arises from two sources. First, households with incorrect beliefs choose suboptimal fertility (relative to a household optimizing under the truth). Malthusians in a Romerian world underinvest in children; Romerians in a Malthusian world overinvest. Second, belief dispersion generates coordination failures even when mean beliefs are correct, because households face strategic uncertainty about others' actions.

To quantify welfare loss, I compare the heterogeneous-belief decentralized equilibrium to a counterfactual *common-belief decentralized equilibrium* in which all households share a common belief $\bar{\theta}$. I emphasize that this counterfactual is not the social planner's first-best, since individual fertility choices do not internalize the scale-effect externality on A'; rather, it is the natural welfare benchmark for evaluating the consequences of belief heterogeneity, holding institutions fixed. Let W^{HB} denote welfare under heterogeneous beliefs and $W(\bar{\theta})$ denote welfare in the common-belief equilibrium with belief $\bar{\theta}$.

Lemma 2 (Welfare Loss from Dispersion)**.** *Suppose the mean belief $\bar{\theta} = \phi\theta_R + (1-\phi)\theta_M$ equals the true parameter θ^*. The welfare loss from belief dispersion, measured relative to the common-belief equilibrium with belief θ^*, is:*

$$W(\theta^*) - W^{HB} = \Omega \cdot Var(\theta_i) + o(Var(\theta_i)), \tag{30}$$

where $\Omega > 0$ is a constant that depends on the curvature of the (true-evaluation) value function and the slope of the fertility policy in beliefs. More generally, the welfare loss equals $\Omega[Var(\theta_i) + (\bar{\theta} - \theta^*)^2] + o(\cdot)$ when mean beliefs are biased.

The proof is in Appendix A. The welfare loss is second-order in belief dispersion, analogous to the welfare cost of price dispersion in New Keynesian models (Woodford, 2003). The loss arises because fertility is a nonlinear function of beliefs, so averaging beliefs before choosing fertility yields different aggregate outcomes than averaging fertility choices across heterogeneous beliefs.

7.3 The Value of a Perfect Announcement

I first consider the case where the planner knows θ^* with certainty and announces it credibly. All households adopt θ^* as their belief, eliminating belief heterogeneity.

Proposition 5 (Value of Perfect Information)**.** *Suppose the planner announces $\theta^A = \theta^*$ and households fully adopt this belief. Then:*

(i) Belief dispersion falls to zero: $Var(\theta_i^{post}) = 0$.

(ii) Welfare increases by $\Omega \cdot [Var(\theta_i) + (\bar{\theta} - \theta^*)^2]$ relative to the heterogeneous-belief equilibrium.

(iii) The fertility gap $n_R - n_M$ falls to zero, and aggregate fertility equals the level chosen in the common-belief decentralized equilibrium with the true parameter (the welfare benchmark in Lemma 2).

The proof follows directly from Lemma 2. Perfect public information eliminates the welfare cost of belief dispersion and attains the common-belief benchmark. (This is the planner's best achievable outcome under information policy alone; closing the residual gap to the social optimum would additionally require correcting the scale-effect externality through, e.g., a fertility subsidy or tax.)

7.4 The Value of an Imperfect Announcement

The more interesting case is when the planner observes only a noisy signal $\hat{\theta} = \theta^* + \nu$. Should the planner announce this imperfect estimate, or remain silent?

Theorem 3 (Value of Imperfect Information)**.** *Suppose the planner announces $\theta^A = \hat{\theta}$ and households fully adopt the announced value (τ_A large), and suppose $\bar{\theta} = \theta^*$ (unbiased prior). Define the value of announcement as:*

$$\mathcal{V}(\tau_\nu) = \mathbb{E}_\nu \left[W^{announce} - W^{HB} \right]. \tag{31}$$

Then, in the local quadratic approximation of Lemma 2:

(i) $\mathcal{V}(\tau_\nu) = \Omega\left(Var(\theta_i) - \tau_\nu^{-1}\right)$, so $\mathcal{V}(\tau_\nu) > 0$ if and only if $\tau_\nu > \underline{\tau} \equiv Var(\theta_i)^{-1}$: the announcement improves expected welfare provided the planner's signal is sufficiently informative relative to belief dispersion.

(ii) $\mathcal{V}(\tau_\nu)$ is strictly increasing in τ_ν: more precise signals yield larger welfare gains.

(iii) In the local approximation, $\mathcal{V}(\tau_\nu)$ is decreasing without bound as τ_ν shrinks under full adoption (forcing households to coordinate on an arbitrarily noisy signal is harmful); under more realistic credibility (τ_A commensurate with τ_ν), households endogenously ignore an uninformative signal, so $\mathcal{V}(\tau_\nu) \to 0$ as $\tau_\nu \to 0$.

(iv) As $\tau_\nu \to \infty$, $\mathcal{V}(\tau_\nu) \to \Omega \cdot Var(\theta_i)$: perfect signals achieve the full welfare gain.

The proof is in Appendix A. The key insight is that public announcements provide two distinct benefits: they move average beliefs toward the truth (if the planner's signal is informative), and they coordinate beliefs by reducing dispersion. When the signal is sufficiently precise, the coordination and information benefits dominate the cost of the residual error ν.

Crucially, when the signal noise exceeds the dispersion of household beliefs, announcement under full adoption can actually *harm* welfare by causing all households to coordinate on a noisy signal, replacing useful private heterogeneity with common error.

Intuition. Consider the boundary case $\tau_\nu = \underline{\tau}$: the variance of the planner's signal equals the variance of household priors. At this point, the gain from eliminating dispersion (which removes $\Omega\text{Var}(\theta_i)$ of loss) is exactly offset in expectation by the new common-error loss ($\Omega\sigma_\nu^2$). For any $\tau_\nu > \underline{\tau}$, the coordination/information benefit strictly dominates, and announcement is beneficial in expectation.

If households update via Bayes' rule with credibility τ_A commensurate with the planner's actual precision, the threshold $\underline{\tau}$ becomes weaker—households endogenously discount uninformative signals. In particular, as $\tau_A \to 0$, households ignore the announcement entirely and $\mathcal{V} \to 0$, so announcement never hurts asymptotically. The full-adoption case in Theorem 3 therefore represents a worst case for the planner; in practice, partial adoption mitigates the downside.

7.5 Decomposing the Welfare Gain

To understand the sources of welfare improvement, I decompose the value of announcement into two components.

Proposition 6 (Decomposition of Announcement Value)**.** *The expected welfare gain from announcement can be written as:*

$$\mathcal{V}(\tau_\nu) = \underbrace{\Omega \cdot \left(Var(\theta_i) - \mathbb{E}[\,Var(\theta_i^{post})]\right)}_{\textit{Coordination gain}} - \underbrace{\Omega \cdot \left(\mathbb{E}[(\bar{\theta}^{post} - \theta^*)^2] - (\bar{\theta} - \theta^*)^2\right)}_{\textit{Information loss/gain}}. \tag{32}$$

The proof is in Appendix A. The first term captures the reduction in belief dispersion, which is non-negative whenever the announcement reduces variance. The second term captures the change in mean-belief error: it is a *gain* if the announcement moves average beliefs closer to the truth, and a *loss* if it moves them away. Whether $\mathcal{V}$ is positive depends on the balance between coordination gain and information loss, which is what produces the precision threshold $\underline{\tau}$ in Theorem 3.

When does coordination dominate? The coordination component is large when initial belief dispersion $\text{Var}(\theta_i)$ is large and when households are highly responsive to the announcement (τ_A large relative to τ_j). In societies with sharp polarization between Romerians and Malthusians, the coordination gain from a common focal point can be substantial.

When does the information component dominate? The information component is large when the planner's signal is precise (τ_ν large) and when prior beliefs are biased ($\bar{\theta} \neq \theta^*$). If households' average belief happens to equal the truth, the expected information component is non-positive, and the planner's case for announcement rests entirely on the coordination component (which must in turn outweigh the expected error introduced by the noisy signal).

7.6 Optimal Announcement Policy

I now consider whether the planner should sometimes remain silent.

Corollary 2 (Optimal Announcement)**.** *Under the conditions of Theorem 3, truthful announcement ($\theta^A = \hat{\theta}$) dominates no announcement whenever $\tau_\nu > \underline{\tau} \equiv Var(\theta_i)^{-1}$. When $\tau_\nu \leq \underline{\tau}$ under full adoption, the planner is better off remaining silent. If credibility is endogenous (τ_A commensurate with τ_ν), truthful announcement weakly dominates silence for any $\tau_\nu > 0$, with strict dominance whenever the signal is informative and beliefs are dispersed.*

This result is consistent with the intuition of Morris and Shin (2002), who show that public information can reduce welfare in pure-complementarity games by inducing excessive coordination on potentially incorrect signals. In my model, the harmful over-coordination is dampened by the heterogeneous strategic interaction: Malthusian strategic substitutability provides a stabilizing counter-response to common signals. As a result, the precision threshold for beneficial announcement ($\underline{\tau} = \mathrm{Var}(\theta_i)^{-1}$) is less stringent than in a homogeneous-complementarity environment, but it does not vanish entirely under full credulity.

Remark 2. *If the population were entirely Romerian ($\phi = 1$), pure strategic complementarity would prevail, and the Morris–Shin concern about over-coordination would be most acute. The presence of Malthusians provides a stabilizing force that lowers the precision threshold required for announcement to improve welfare.*

7.7 Policy Implications

The analysis yields several implications for government communication policy regarding long-run economic relationships.

Announce when the signal is informative; remain silent otherwise. Governments should communicate their best estimates of long-run relationships when they have meaningful information (τ_ν above the dispersion-based threshold). Conversely, broadcasting a near-uninformative estimate as if it were authoritative can be harmful by causing the population to coordinate on noise.

Credibility matters. The welfare gains from announcement depend on households treating the announcement as informative. Governments that have historically provided reliable economic information will achieve larger coordination gains. Endogenous credibility—in which households appropriately discount low-precision signals—also serves as insurance against announcement errors.

Reducing polarization has intrinsic value. Beyond the direct welfare effects of fertility choices, belief dispersion is costly because it undermines coordination. Policies that reduce polarization—even if they do not change average beliefs—can improve welfare by facilitating coordination.

The optimal message depends on the audience. In a population heavily dominated by Romerians, the planner should communicate cautiously to avoid over-coordination. In a population with substantial Malthusian presence, the planner can communicate more aggressively because Malthusian strategic substitutability provides a natural stabilizer.

7.8 Quantifying the Welfare Gain

To provide a sense of magnitudes, I calibrate the welfare gain from announcement using illustrative parameters.

Suppose $\theta_R = 0.1$, $\theta_M = -0.1$, and $\phi = 0.5$ (equal population shares). The belief variance is $\mathrm{Var}(\theta_i) = 0.5 \times (0.1)^2 + 0.5 \times (-0.1)^2 = 0.01$.

With $\Omega \approx 0.5$ (calibrated from the curvature of utility and fertility functions), the welfare loss from belief dispersion is approximately $0.5 \times 0.01 = 0.005$, or 0.5% of steady-state welfare. The dispersion-based precision threshold is $\underline{\tau} = 1/0.01 = 100$, corresponding to $\sigma_\nu^2 = 0.01$ (a standard error of 0.1 in estimating θ).

A planner with signal precision $\tau_\nu = 400$ (standard error of 0.05 in estimating θ, well below the threshold of 0.1) achieves $\mathcal{V} = \Omega(\mathrm{Var}(\theta_i) - \tau_\nu^{-1}) = 0.5 \times (0.01 - 0.0025) = 0.00375$, or roughly 0.375% of steady-state welfare. In consumption-equivalent terms, this is comparable to the welfare gains from moderate reductions in business cycle volatility.

8 Conclusion

This paper has developed a theory of fertility choice under heterogeneous beliefs about population scale effects. The main contributions are fourfold.

First, I show that, provided the Romerian scale-effect belief is large enough to overcome the dilution of per capita consumption, Romerians (who believe population enhances productivity) exhibit strategic complementarity in fertility choices, while Malthusians (who believe population diminishes productivity) always exhibit strategic substitutability. This asymmetry has important implications for equilibrium multiplicity and stability.

Second, I characterize long-run dynamics under demographic selection and Bayesian learning. In Romerian worlds, both forces align, leading to convergence and efficiency. In Malthusian worlds, the forces conflict, generating either persistent inefficiency or endogenous boom-bust cycles depending on the relative strength of learning versus demographic selection.

Third, I apply the model to interpret demographic history. The Malthusian cycles of the pre-industrial era, the demographic transition accompanying industrialization, and the modern fertility decline can all be understood through the lens of belief heterogeneity and its interaction with the true population-productivity relationship.

Fourth, I analyze the value of public information. When a social planner's signal is sufficiently informative relative to the dispersion of household beliefs, announcing an estimate improves welfare by coordinating beliefs and reducing fertility dispersion. When the signal is too noisy and households fully adopt it, announcement can backfire by inducing coordination on noise; under endogenous credibility, this downside is mitigated because households discount low-precision signals.

The empirical analysis using General Social Survey data provides suggestive support for the model's central mechanism. Individuals who hold Malthusian beliefs—agreeing that Earth cannot continue to support population growth—report ideal fertility approximately 0.27 children lower and have 0.29 fewer actual children than non-Malthusians, differences of roughly 11 and 16 percent respectively. These relationships are robust to controls for age, education, and sex, hold across demographic groups, and persist among those with completed fertility. The fact that beliefs correlate with realized fertility outcomes, not just stated preferences, strengthens the case that beliefs shape consequential behavior. While the evidence is correlational and cannot definitively establish causality, it is consistent with the prediction that beliefs about population-productivity relationships shape fertility decisions.

Several extensions merit future research. First, I have treated the distribution of beliefs as evolving through cultural transmission and learning; endogenizing the intensity of cultural transmission as a parental choice would enrich the model. Second, incorporating human capital and the quantity-quality tradeoff would allow for a richer analysis of the demographic transition. Third, extending the model to an open economy with migration would introduce additional channels through which beliefs affect population dynamics. Fourth, developing credible identification strategies to establish the causal effect of beliefs on fertility—perhaps through survey experiments or natural experiments in exposure to Malthusian ideas—would strengthen the empirical foundation.

From a policy perspective, the model suggests that governments should communicate their best estimates of long-run economic relationships when they have substantive information, while exercising restraint when uncertainty is large enough to risk harmful over-coordination. More broadly, reducing belief polarization has intrinsic value beyond its effects on average beliefs, because dispersion itself undermines coordination and efficiency.

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A Proofs

A.1 Proof of Lemma 1

I solve the household optimization problem and verify the value function conjecture.

Step 1: Bellman equation. The Bellman equation for type j is:

$$V_j(A, N, \phi) = \max_{n \geq 0} \left\{ \ln(w - \psi(n)) + \beta n \cdot \mathbb{E}_j[V_j(A', N', \phi')] \right\}, \tag{33}$$

where $w = \alpha A N^{\alpha-1}$ and expectations are taken under belief θ_j.

Step 2: Conjecture. Conjecture $V_j(A, N, \phi) = v_j(\phi) + \xi_{A,j} \ln A + \xi_{N,j} \ln N$, where the constants $\xi_{A,j}, \xi_{N,j}$ are allowed to depend on type and on the equilibrium fertility n_j^*. The flow utility $\ln(w - \psi(n)) = \ln \alpha + \ln A + (\alpha - 1) \ln N + \ln(1 - \psi(n)/w)$ admits a log-linear part plus a residual term $\ln(1 - \psi(n)/w) \approx -\psi(n)/w$ that contributes only to $v_j(\phi)$ at leading order.

Step 3: Expected continuation value. Under the laws of motion (taking the log-linear approximation $\ln(1 + x) \approx x$ for small x):

$$\ln A' \approx \ln A + \bar{g} + \theta_j \ln N, \tag{34}$$

$$\ln N' = \ln N + \ln \bar{n}. \tag{35}$$

Thus

$$\begin{aligned} \mathbb{E}_j[\xi_{A,j} \ln A' + \xi_{N,j} \ln N'] = \xi_{A,j} \ln A + \xi_{N,j} \ln N& \\ + \xi_{A,j} \bar{g} + \xi_{A,j} \theta_j \ln N + \xi_{N,j} \ln \bar{n}&. \end{aligned} \tag{36}$$

Step 4: Matching coefficients. At the optimal n_j^*, matching coefficients of $\ln A$ and $\ln N$ on both sides of the Bellman equation gives:

$$\xi_{A,j} = 1 + \beta n_j^* \xi_{A,j} \quad \Rightarrow \quad \xi_{A,j} = \frac{1}{1 - \beta n_j^*}, \tag{37}$$

$$\xi_{N,j} = (\alpha - 1) + \beta n_j^* \big(\xi_{A,j} \theta_j + \xi_{N,j} \big) \quad \Rightarrow \quad \xi_{N,j} = \frac{(\alpha - 1)(1 - \beta n_j^*) + \beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}. \tag{38}$$

Both coefficients depend on the type-specific equilibrium fertility n_j^* and (in the case of $\xi_{N,j}$) explicitly on θ_j. The standing assumption $\beta n_j^* < 1$ ensures finiteness.

Step 5: First-order condition. With the quadratic cost $\psi(n) = \frac{\gamma}{2}n^2$, the FOC is:

$$\frac{\gamma n}{w - \frac{\gamma}{2}n^2} = \beta \mathbb{E}_j[V_j(A', N', \phi')]. \tag{39}$$

The right-hand side is a strictly increasing function of θ_j (through both $\xi_{A,j}\theta_j \ln N$ and the increase of $\xi_{N,j}$ in θ_j, holding equilibrium objects fixed at their values), and the left-hand side is strictly increasing in n. Therefore the implicit solution n_j^* is strictly increasing in θ_j, and we can write $n_j^* = n^* \cdot h(\theta_j; \bar{n})$ for a baseline level n^* and a function h that is increasing in its first argument.

Step 6: Fertility ranking. Since h is increasing and $\theta_R > \theta_M$, we have $n_R^* > n_M^*$. $\square$

A.2 Proof of Proposition 1

I derive the dependence of optimal fertility on expected aggregate fertility.

Step 1: Setup. Consider type j's problem when the household expects aggregate fertility $\bar{n}^e$. The continuation value is

$$\mathbb{E}_j[V_j(A', N', \phi')] = v_j(\phi') + \xi_{A,j}[\ln A + \bar{g} + \theta_j \ln N] + \xi_{N,j}[\ln N + \ln \bar{n}^e], \tag{40}$$

where $\xi_{A,j}$ and $\xi_{N,j}$ are as derived in the proof of Lemma 1. Crucially, current-period TFP growth depends on current population N, not on $\bar{n}^e$; the dominant direct dependence of the continuation value on $\bar{n}^e$ comes through $N' = \bar{n}^e N$, and is captured by the coefficient $\xi_{N,j}$ (which encapsulates the discounted infinite stream of future scale-effect and dilution effects). The indirect channel through ϕ' enters $v_j(\phi')$ and is of higher order for the local sign analysis.

Step 2: Sign of the marginal continuation value. Differentiating,

$$\frac{\partial \mathbb{E}_j[V_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}, \tag{41}$$

and from Step 4 of the previous proof, the sign of $\xi_{N,j}$ equals the sign of

$$(\alpha - 1)(1 - \beta n_j^*) + \beta n_j^* \theta_j. \tag{42}$$

For Malthusians ($\theta_M < 0$): both $(\alpha - 1)(1 - \beta n_M^*) < 0$ and $\beta n_M^* \theta_M < 0$, so $\xi_{N,M} < 0$ unambiguously.

For Romerians ($\theta_R > 0$): the second term is positive but the first is negative; $\xi_{N,R} > 0$ iff $\theta_R > (1-\alpha)(1-\beta n_R^*)/(\beta n_R^*)$. This is condition (21), which is assumed.

Step 3: Effect on optimal fertility. From the FOC,

$$\frac{\gamma n_j}{w - \frac{\gamma}{2} n_j^2} = \beta \mathbb{E}_j[V_j], \tag{43}$$

the left-hand side is strictly increasing in n_j, so $\text{sign}(dn_j^*/d\bar{n}^e) = \text{sign}(\partial \mathbb{E}_j[V_j]/\partial \bar{n}^e) = \text{sign}(\xi_{N,j})$.

Combining with Step 2: $dn_R^*/d\bar{n}^e > 0$ (strategic complementarity for Romerians, under (21)) and $dn_M^*/d\bar{n}^e < 0$ (strategic substitutability for Malthusians). $\square$

A.3 Proof of Proposition 2

Step 1: Aggregate best response. Define $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1-\phi) n_M^*(\bar{n}^e)$.

Step 2: Slope.

$$\frac{dBR}{d\bar{n}^e} = \phi \frac{dn_R^*}{d\bar{n}^e} + (1-\phi) \frac{dn_M^*}{d\bar{n}^e}. \tag{44}$$

From Proposition 1, $dn_R^*/d\bar{n}^e > 0$ and $dn_M^*/d\bar{n}^e < 0$. The slope is positive when ϕ is large, negative when ϕ is small, and zero at

$$\phi^* = \frac{|dn_M^*/d\bar{n}^e|}{dn_R^*/d\bar{n}^e + |dn_M^*/d\bar{n}^e|}. \tag{45}$$

Step 3: Uniqueness condition. An equilibrium requires $\bar{n}^e = BR(\bar{n}^e)$. Graphically, this is the intersection of the 45-degree line with $BR(\bar{n}^e)$.

When $\phi < \phi^*$, BR slopes downward. Since $BR(0) > 0$ and $BR(\bar{n}^e)$ is bounded, there is exactly one intersection: unique equilibrium.

When $\phi > \phi^*$, BR slopes upward. If the slope exceeds 1 at some point, the curves can intersect multiple times, generating multiple equilibria.

Step 4: Ranking equilibria. When multiple equilibria exist, they can be ordered. Let $\bar{n}^L < \bar{n}^M < \bar{n}^H$ be three equilibria (if they exist). Standard arguments show that $\bar{n}^L$ and $\bar{n}^H$ are stable under best-response dynamics while $\bar{n}^M$ is unstable. $\square$

A.4 Proof of Proposition 3

The proof adapts the global games equilibrium selection argument of Morris and Shin (1998) to our setting with heterogeneous strategic interaction.

Step 1: Setup with private signals. Each household observes a private signal $\phi_i = \phi + \eta_i$ where $\eta_i \sim N(0, \sigma_\eta^2)$ i.i.d. The true ϕ is drawn from a prior; for simplicity, assume $\phi \sim U[0,1]$. We restrict attention to the parameter region of Proposition 2(ii), in which the complete-information game admits two stable equilibria, naturally labeled "low" and "high" by aggregate fertility.

Step 2: Threshold strategies. Consider symmetric threshold strategies in which each type's equilibrium choice in the continuous-action game is selected by comparing the private signal to a threshold:

- Romerians: n_R^H if $\phi_i > \phi^{**}$, n_R^L if $\phi_i < \phi^{**}$;
- Malthusians: n_M^L if $\phi_i > \phi^{**}$, n_M^H if $\phi_i < \phi^{**}$.

The Malthusian strategy is reversed because of strategic substitutability: when Malthusians infer that the Romerian share is high (and hence aggregate fertility is high), they reduce their own fertility.

Step 3: Indifference condition. At the threshold ϕ^{**}, each type must be indifferent between the high and low equilibrium choices. This requires that the expected aggregate fertility implied by signal ϕ^{**} makes both types simultaneously indifferent at ϕ^{**}.

For a household of type j with signal $\phi_i = \phi^{**}$, the expected aggregate fertility integrates over the posterior on ϕ and over the resulting fractions of agents above and below the threshold. As $\sigma_\eta \to 0$, ϕ_i becomes increasingly informative about ϕ, and the conditional distribution of the rank of ϕ^{**} within the population converges to the uniform Laplacian distribution on $[0,1]$, the standard global-games limit.

Step 4: Fixed point. The threshold ϕ^{**} is determined by a fixed-point condition combining the Romerian and Malthusian indifference equations. Because the Romerian indifference equation is monotone in one direction and the Malthusian indifference equation is monotone in the other direction (a consequence of opposite strategic interactions), the fixed point is unique provided both indifference functions are continuous and intersect—which they do under standard regularity.

Step 5: Uniqueness. By the standard global games iterated dominance argument applied to the binary equilibrium-selection game, as $\sigma_\eta \to 0$ the unique threshold equilibrium is selected. This pins down a unique outcome from among the multiple equilibria that exist under common knowledge. $\square$

A.5 Proof of Theorem 1

I show convergence in Romerian worlds using a direct Bayesian-consistency argument combined with a deterministic monotonicity argument for demographic selection.

Step 1: Dynamics. The Romerian share evolves as:

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1-\lambda)\pi_{t+1}, \tag{46}$$

where $\phi_t^{birth} = \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t) n_M}$ and π_{t+1} is the Bayesian posterior.

Step 2: Demographic selection. For $\phi_t \in (0,1)$, since $n_R > n_M$:

$$\phi_t^{birth} > \phi_t \iff \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t)n_M} > \phi_t \iff n_R > \phi_t n_R + (1-\phi_t) n_M, \tag{47}$$

and the last inequality reduces to $(1-\phi_t)(n_R - n_M) > 0$, which holds. Thus $\phi_t^{birth} > \phi_t$ strictly for any interior ϕ_t.

Step 3: Learning. In a Romerian world ($\theta^* = \theta_R$), the signal is $s_t = \bar{g} + \theta_R \ln N_t + \varepsilon_t$. The log-likelihood ratio satisfies

$$\ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)} = \frac{(\theta_R - \theta_M)\ln N_t \cdot \left[2(s_t - \bar{g}) - (\theta_R + \theta_M)\ln N_t\right]}{2\sigma_\varepsilon^2}, \tag{48}$$

and substituting $s_t - \bar{g} = \theta_R \ln N_t + \varepsilon_t$ and taking expectations under the true Romerian model:

$$\mathbb{E}^{\theta_R}\left[\ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)} \,\Big|\, \mathcal{F}_{t-1}\right] = \frac{(\theta_R - \theta_M)^2 (\ln N_t)^2}{2\sigma_\varepsilon^2} > 0 \tag{49}$$

whenever $\ln N_t \neq 0$. Under the true Romerian measure, the process $\{\pi_t\}$ is a bounded $[0,1]$-valued submartingale: writing the Bayesian update as $\pi_{t+1} = \pi_t \ell_R / [\pi_t \ell_R + (1-\pi_t)\ell_M]$, one has $\mathbb{E}^{\theta_R}[\pi_{t+1} \mid \mathcal{F}_t] \geq \pi_t$ (by Jensen's inequality applied to the convex function $x \mapsto x/[x + (1-\pi_t)/\pi_t \cdot \ell_M/\ell_R]$ of the likelihood ratio, whose mean exceeds 1). By the submartingale convergence theorem, π_t converges almost surely to a limit $\pi_\infty \in [0,1]$. Strict positivity of the expected log-likelihood ratio whenever $\pi_t \in (0,1)$ and $\ln N_t \neq 0$ rules out an interior limit (this is the standard Bayesian consistency / Doob's a.s. identification theorem). Hence $\pi_t \to 1$ almost surely.

Step 4: Combined dynamics. For any interior (ϕ_t, π_t), $\phi_t^{birth} > \phi_t$ deterministically, and $\pi_t \to 1$ a.s. by Step 3. From (13),

$$1 - \phi_{t+1} = \lambda(1 - \phi_t^{birth}) + (1 - \lambda)(1 - \pi_{t+1}) \leq \lambda(1 - \phi_t) + (1 - \lambda)(1 - \pi_{t+1}).$$

For any $\varepsilon > 0$, eventually $1 - \pi_{t+1} < \varepsilon$ a.s., so iterating the inequality gives $\limsup_{k\to\infty}(1 - \phi_{t+k}) \leq \varepsilon/(1 - \lambda)$ a.s. (or $\leq \varepsilon$ when $\lambda < 1$ suitably bounded). Since ε is arbitrary, $\phi_t \to 1$ almost surely.

Step 5: Conclusion. Aggregate fertility $\bar{n}_t = \phi_t n_R + (1 - \phi_t)n_M$ converges to n_R^*, and population grows at rate $n_R^* - 1$ in the long run. □

A.6 Proof of Theorem 2

I establish the existence of cycles using local bifurcation analysis around an interior steady state.

Step 1: Linearized dynamics. Consider the system in $(\phi_t, \ln N_t)$. Let $\hat{\phi}_t = \phi_t - \phi^{ss}$ and $\hat{n}_t = \ln N_t - \ln N^{ss}$ denote deviations from an interior steady state of the deterministic skeleton.

The linearized dynamics are:

$$\begin{pmatrix} \hat{\phi}_{t+1} \\ \hat{n}_{t+1} \end{pmatrix} = J \begin{pmatrix} \hat{\phi}_t \\ \hat{n}_t \end{pmatrix}, \tag{50}$$

where J is the Jacobian matrix evaluated at the steady state.

Step 2: Jacobian entries. The entries of J are:

- $J_{11} = \partial\phi_{t+1}/\partial\phi_t > 0$: more Romerians today mean more Romerian children through demographic selection.
- $J_{12} = \partial\phi_{t+1}/\partial \ln N_t < 0$ in a Malthusian world: higher N_t leads to lower realized productivity growth, which updates beliefs toward Malthusianism.
- $J_{21} = \partial \ln N_{t+1}/\partial\phi_t > 0$: more Romerians mean higher aggregate fertility.

- $J_{22} = \partial \ln N_{t+1}/\partial \ln N_t = 1 + \partial \ln \bar{n}_t/\partial \ln N_t$; the sign of the second term depends on how fertility responds to wages and is generally ambiguous but bounded.

Step 3: Eigenvalue analysis. The characteristic polynomial is

$$\mu^2 - \tau\mu + \Delta = 0, \tag{51}$$

with trace $\tau = J_{11} + J_{22}$ and determinant $\Delta = J_{11}J_{22} - J_{12}J_{21}$. Eigenvalues are complex when $\tau^2 < 4\Delta$; when complex, their modulus equals $\sqrt{\Delta}$, so the modulus exceeds one iff $\Delta > 1$.

Step 4: Conditions for oscillatory instability. Since $J_{12} < 0$ and $J_{21} > 0$ in a Malthusian world, the cross-term $-J_{12}J_{21} > 0$ raises Δ. For sufficiently strong learning ($|J_{12}|$ large enough that $\Delta > 1$) but not too strong ($\tau^2 < 4\Delta$), we obtain complex eigenvalues with $|\mu| > 1$.

Step 5: Limit cycle. When learning is very weak ($|J_{12}|$ small), $\Delta < 1$ and eigenvalues are real with modulus ≤ 1; the interior steady state is locally stable and the system converges monotonically. When learning is very strong ($|J_{12}|$ very large), the interior steady state ceases to exist or shifts toward a corner with Malthusian dominance, where Malthusian beliefs prevail; the relevant dynamics are now around that corner, and oscillations cease. For intermediate $|J_{12}|$, the linearized system around the interior steady state is locally explosive with complex eigenvalues. The global nonlinear system is bounded (since $\phi \in [0, 1]$ and population is bounded by carrying capacity), so the explosive linear dynamics translate into bounded oscillations consistent with a stable limit cycle. The standard Neimark–Sacker bifurcation argument then establishes the existence of an attracting invariant closed curve when $|\mu|$ crosses one transversally. □

A.7 Proof of Proposition 4

Step 1: Define indices. Let $\mathcal{L} = \sigma_\varepsilon^{-2}|\theta_R - \theta_M|$ index learning responsiveness and $\mathcal{D} = \lambda(n_R - n_M)/\bar{n}$ index demographic selection.

Step 2: Thresholds. From the eigenvalue analysis in the proof of Theorem 2:

- $\underline{\mathcal{L}}(\mathcal{D})$ is the smallest $\mathcal{L}$ such that $\Delta > 1$ holds; below this, demographic selection dominates and $\phi \to 1$ monotonically.

- $\overline{\mathcal{L}}(\mathcal{D})$ is the largest $\mathcal{L}$ consistent with the existence of an interior steady state and $\tau^2 < 4\Delta$; above this, the relevant fixed point migrates to a corner and oscillation around the interior point ceases.

Step 3: Derivation. Setting $\Delta = 1$ and $\tau^2 = 4\Delta$ yields two equations in the entries of J, which are themselves functions of $\mathcal{L}$ and $\mathcal{D}$. Solving for $\mathcal{L}$ at each boundary gives the threshold functions. The algebra is standard but tedious and is omitted. □

A.8 Proof of Lemma 2

Step 1: Setup. Aggregate welfare under heterogeneous beliefs is

$$W^{HB} = \phi V_R + (1-\phi)V_M, \tag{52}$$

where V_j is the (true) value attained by a type-j household choosing fertility n_j^* under belief θ_j, evaluated under the true law of motion. The benchmark welfare $W(\theta^*)$ is the corresponding object in the counterfactual common-belief decentralized equilibrium where every household holds the correct belief θ^* and chooses the resulting individually-optimal fertility n^{CB}. I emphasize that $W(\theta^*)$ is not the social planner's first-best, since individual choices do not internalize the scale-effect externality on A'.

Step 2: Quadratic approximation. Let n^{CB} denote the common-belief equilibrium fertility—the fertility level a household would choose under the correct belief θ^*, holding aggregates fixed at their common-belief equilibrium values. By the envelope theorem applied to the household's correct-belief problem, the (true-evaluation) value function $V(n)$ viewed as a function of own fertility n satisfies $V'(n^{CB}) = 0$. Standard Taylor expansion then gives

$$V^* - V_j \approx \frac{\omega}{2}(n_j^* - n^{CB})^2, \tag{53}$$

where $V^* = V(n^{CB})$ and $\omega \equiv -V''(n^{CB}) > 0$ is the curvature of the (true-evaluation) value at n^{CB}.

Step 3: Linking fertility to beliefs. Linearizing the policy function around θ^*, $n_j^* - n^{CB} \approx \kappa(\theta_j - \theta^*) + o(\theta_j - \theta^*)$, where $\kappa > 0$ is the slope of the fertility policy in beliefs.

Step 4: Aggregate loss. Combining the previous two steps,

$$\begin{aligned} W(\theta^*) - W^{HB} &= \phi(V^* - V_R) + (1-\phi)(V^* - V_M) \\ &\approx \frac{\omega\kappa^2}{2}\left[\phi(\theta_R - \theta^*)^2 + (1-\phi)(\theta_M - \theta^*)^2\right] \quad (54) \\ &= \frac{\omega\kappa^2}{2}\left[\mathrm{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2\right], \quad (55) \end{aligned}$$

which equals $\Omega \cdot \mathrm{Var}(\theta_i)$ when $\bar{\theta} = \theta^*$, with $\Omega = \omega\kappa^2/2$. □

A.9 Proof of Theorem 3

Step 1: Post-announcement beliefs. Under full adoption (τ_A large), households' posterior belief equals the announced value $\theta^A = \theta^* + \nu$. Therefore $\mathrm{Var}(\theta_i^{post}) = 0$, but $\bar{\theta}^{post} = \theta^* + \nu$, so the post-announcement mean belief differs from the truth by ν.

Step 2: Welfare with announcement. Applying Lemma 2 to the post-announcement belief distribution,

$$W(\theta^*) - W^{announce} = \Omega\left[\mathrm{Var}(\theta_i^{post}) + (\bar{\theta}^{post} - \theta^*)^2\right] = \Omega\nu^2. \quad (56)$$

Step 3: Welfare without announcement. Under $\bar{\theta} = \theta^*$, Lemma 2 gives

$$W(\theta^*) - W^{HB} = \Omega \mathrm{Var}(\theta_i). \quad (57)$$

Step 4: Expected gain. Subtracting,

$$\begin{aligned} \mathcal{V}(\tau_\nu) &= \mathbb{E}_\nu\left[W^{announce} - W^{HB}\right] = \Omega \mathrm{Var}(\theta_i) - \Omega \mathbb{E}[\nu^2] \\ &= \Omega\big(\mathrm{Var}(\theta_i) - \sigma_\nu^2\big) = \Omega\big(\mathrm{Var}(\theta_i) - \tau_\nu^{-1}\big). \quad (58) \end{aligned}$$

Step 5: Properties. Within the local quadratic approximation:

- $\mathcal{V}(\tau_\nu) > 0 \iff \tau_\nu > \underline{\tau} \equiv \mathrm{Var}(\theta_i)^{-1}$. Hence part (i).
- $\mathcal{V}'(\tau_\nu) = \Omega\tau_\nu^{-2} > 0$. Hence part (ii).
- Under full adoption, $\mathcal{V} \to -\infty$ as $\tau_\nu \to 0$ within the local approximation: forcing households to coordinate on a noisier and noisier signal is increasingly harmful. Under endogenous credibility (τ_A commensurate with τ_ν), Bayesian updating gives $\theta_j^{post} =$

$(\tau_j \theta_j + \tau_\nu \theta^A)/(\tau_j + \tau_\nu) \to \theta_j$ as $\tau_\nu \to 0$, so households ignore the announcement and $\mathcal{V} \to 0$. This establishes part (iii).

- As $\tau_\nu \to \infty$, $\mathcal{V} \to \Omega \mathrm{Var}(\theta_i)$, the full first-best dispersion gain. Hence part (iv). $\square$

A.10 Proof of Proposition 6

Step 1: Welfare difference. From Lemma 2, the welfare loss with and without announcement is, respectively,

$$W(\theta^*) - W^{announce} = \Omega\big[\mathrm{Var}(\theta_i^{post}) + (\bar{\theta}^{post} - \theta^*)^2\big], \tag{59}$$

$$W(\theta^*) - W^{HB} = \Omega\big[\mathrm{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2\big]. \tag{60}$$

Step 2: Differencing.

$$W^{announce} - W^{HB} = \Omega\big[\mathrm{Var}(\theta_i) - \mathrm{Var}(\theta_i^{post})\big] - \Omega\big[(\bar{\theta}^{post} - \theta^*)^2 - (\bar{\theta} - \theta^*)^2\big]. \tag{61}$$

Step 3: Taking expectations.

$$\mathcal{V}(\tau_\nu) = \Omega\big(\mathrm{Var}(\theta_i) - \mathbb{E}[\mathrm{Var}(\theta_i^{post})]\big) - \Omega\big(\mathbb{E}[(\bar{\theta}^{post} - \theta^*)^2] - (\bar{\theta} - \theta^*)^2\big). \tag{62}$$

The first bracket, the *coordination gain*, is non-negative whenever the announcement reduces dispersion. The second bracket, the *information loss/gain*, is positive (a loss) when the announcement moves average beliefs away from the truth in expectation, and negative (a gain) when it moves them closer. Whether $\mathcal{V} > 0$ depends on the balance between these terms, which is exactly the precision threshold derived in Theorem 3. $\square$